

2025-26

Time - 3 hours

Full Marks - 100

Answer all parts as per instructions.

Parts of each question should be answered continuously.

Figures in the right hand margin indicate marks.

PART – I

1. Answer all questions by choosing the correct answer from the given alternatives. [1 × 10]

(a) Which component of CPU performs arithmetic and logical operations ?

- | | |
|----------------|------------|
| (i) CU | (ii) ALU |
| (iii) Register | (iv) Cache |

(b) Which memory retains data even after power is switched off ?

- | | |
|-----------|----------------|
| (i) RAM | (ii) Cache |
| (iii) ROM | (iv) Registers |

(c) Which Python data type stores elements as key-value pairs ?

- | | |
|-----------|-----------------|
| (i) List | (ii) Tuple |
| (iii) Set | (iv) Dictionary |

P.T.O.

[2]

(d) Output of the following code is :

```
print(4 + 3 * 2 ** 2)
```

- (i) 28 (ii) 16
- (iii) 10 (iv) 20

(e) Which network device connects two similar networks ?

- (i) Router (ii) Hub
- (iii) Switch (iv) Bridge

(f) Which keyword is used to skip the current iteration of a loop ?

- (i) break (ii) pass
- (iii) continue (iv) exit

(g) What will be the output ?

```
print(type((5)))
```

- (i) tuple (ii) int
- (iii) list (iv) error

(h) Which of the following is an immutable data type ?

- (i) List (ii) Dictionary
- (iii) Set (iv) Tuple

(i) What does the len() function return ?

- (i) Size of memory

[3]

(ii) Number of elements

(iii) Data type

(iv) Index value

(j) Output of the code :

```
print("hello".upper())
```

- (i) hello (ii) Hello
- (iii) HELLO (iv) Error

PART – II

2. Answer all questions within 50 words each. [2 × 9]

- (a) Define application software with two examples.
- (b) Differentiate between LAN and WAN.
- (c) What is an algorithm ? Mention two characteristics.
- (d) Explain mutable and immutable objects with examples.
- (e) Write a Python program to check whether a number is prime.
- (f) Correct the code :

```
x = input(Enter number)
```

```
if x > 0
```

```
print("Positive")
```


[4]

- (g) What is the purpose of indentation in Python ?
- (h) Write two differences between list and set.
- (i) Write a Python program to check whether a number is prime.

PART – III

3. Answer any eight of the following questions within 250 words each. [5 × 8]

- (a) Explain different generations of computers.
- (b) Describe types of computer networks with applications.
- (c) Explain types of programming errors with examples.
- (d) Write a Python program to find the largest of three numbers.
- (e) Explain string slicing with examples.
- (f) Write a Python program to check whether a number is palindrome.
- (g) Explain lists and their operations with examples.
- (h) Write a Python program to generate Fibonacci series using loop.

[5]

- (i) Correct the following program :

```
1st = [10, 20, 30]
```

```
for i in range(0, 4):
```

```
    print(1st[i])
```

- (j) Write a Python program to count vowels in a string.

PART – IV

Answer any four questions within 800 words each.

- 4. Explain memory hierarchy with diagram and characteristics. [8]
- 5. Discuss system software, application software, and utility software with examples. [8]
- 6. Explain loops in Python with suitable programs. [8]
- 7. Write a program that : [8]
 - (a) Read N numbers from the user and store them in a list.
 - (b) Separate the numbers into two different lists : one containing even numbers and the other containing odd numbers. Display both the even and odd number lists.
 - (c) Find and display the total count of even numbers and odd numbers.

P.T.O

[6]

(d) Find and display the sum of all even numbers and the sum of all odd numbers.

8. Explain different types of function arguments with examples.

[8]